# ANATOMY OF E

Ophthalmology | 3rd Year MBBS

Hi Sammy!
Anatomy = foundation of
ALL ophthalmology!
Learn this FIRST!

Anatomy of Eye | Ophthalmology | 3rd Year | MBBS

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# Layers of the Eyeball + Dimensions

3 coats of the eyeball
= most basic question!
Know them inside out!

DIMENSIONS (Adult Eyeball)

AP: 24mm | Horizontal: 23.5mm | Vertical: 23mm

Circumference: 75mm | Volume: 6.5ml | Weight: 7g

## Three Coats (Tunics) of the Eyeball

| Coat | Parts | Key Features |
|---|---|---|
| **Outer (Fibrous)** | Sclera (5/6) + Cornea (1/6) | Protective, maintains shape, junction = limbus |
| **Middle (Vascular)** | Iris + Ciliary body + Choroid | Uveal tract — nutrition + accommodation + pupil |
| **Inner (Nervous)** | Retina (10 layers) | Photoreceptors (rods/cones), macula, optic disc |

## Sclera — Key Points

| Feature | Detail |
|---|---|
| **Composition** | Dense collagen + elastic fibers (avascular) |
| **Thinnest** | Behind insertion of recti muscles (0.3mm) |
| **Thickest** | Posterior pole near optic nerve (1mm) |
| **Lamina cribrosa** | Sieve-like area — optic nerve exits through it |
| **Schlemm's canal** | At limbus — drains aqueous humor |
| **Episclera** | Loose connective tissue on outer surface |

## Uveal Tract (Middle Coat)

| Part | Function | Key Feature |
|---|---|---|
| **Iris** | Controls pupil size | Sphincter (para) + Dilator (sympa) |
| **Ciliary body** | Accommodation + aqueous | Ciliary muscle + ciliary processes |
| **Choroid** | Nutrition to outer retina | Most vascular tissue in body |

★ Retina — 10 Layers Mnemonic

**"I Play Extremely Good Innings, I Never Gave Losers Runs"**

ILM, NFL, GCL, IPL, INL, OPL, ONL, ELM, Rods & Cones, RPE

Cornea layers and lens anatomy — asked in EVERY theory AND viva!

## Cornea — 5 Layers (Anterior to Posterior)

| Layer | Type | Key Feature |
|---|---|---|
| **Epithelium** | Stratified squamous | 5-6 layers, regenerates, no scarring |
| **Bowman's membrane** | Acellular collagen | Does NOT regenerate once damaged |
| **Stroma** | Collagen lamellae | 90% of corneal thickness, keratocytes |
| **Descemet's membrane** | Basement membrane | Strong, elastic, regenerates |
| **Endothelium** | Single cell layer | Pump function, does NOT regenerate |

- Avascular (nutrition from aqueous + tears + limbal vessels) | Refractive power: +43D
- Transparency due to: uniform collagen spacing, relative dehydration, avascularity
- Sensory supply: ophthalmic division of trigeminal nerve (long ciliary nerves)

## Crystalline Lens

| Feature | Detail |
|---|---|
| **Shape** | Biconvex, transparent, avascular, no nerve supply |
| **Refractive power** | +15 to +20 D (changes with accommodation) |
| **Capsule** | Thickest basement membrane in body, elastic |
| **Zonules of Zinn** | Suspend lens from ciliary body |
| **Lens proteins** | Crystallins (alpha, beta, gamma) |
| **Growth** | Continues throughout life (new fibers added) |

## Chambers + Aqueous Humor

| Structure | Detail |
|---|---|
| **Anterior chamber** | Between cornea & iris (depth ~2.5mm, volume ~0.25ml) |
| **Posterior chamber** | Between iris & lens (contains aqueous) |
| **Aqueous production** | Ciliary epithelium (active secretion + ultrafiltration) |
| **Aqueous drainage** | Trabecular meshwork -> Schlemm's canal -> episcleral veins |
| **Vitreous humor** | 99% water, fills 4/5th of globe, volume ~4ml |

**Cornea Layers Mnemonic**
"EBSDE" = Epithelium, Bowman, Stroma, Descemet, Endothelium (ant -> post)

# Blood Supply + Nerve Supply

Blood supply of eye is
a classic LAQ topic!
Draw diagrams in exam!

## Arterial Supply (from Ophthalmic Artery)

| Artery | Supplies |
|---|---|
| **Central retinal artery** | Inner 2/3 of retina (END artery — no anastomoses!) |
| **Short posterior ciliary** | Choroid, outer 1/3 retina, circle of Zinn-Haller |
| **Long posterior ciliary** | Major arterial circle of iris (2 arteries) |
| **Anterior ciliary** | From recti muscles -> major arterial circle |
| **Lacrimal artery** | Lacrimal gland, eyelids |
| **Supraorbital artery** | Forehead, scalp |

★ **Central retinal artery = END ARTERY. Occlusion -> sudden painless vision loss!**
Cherry red spot at macula (supplied by choroid). EMERGENCY — treat within 90 min!

## Venous Drainage

| Vein | Drains Into |
|---|---|
| **Central retinal vein** | Superior ophthalmic vein / cavernous sinus |
| **Vortex veins (4-8)** | Superior & inferior ophthalmic veins |
| **Anterior ciliary veins** | Episcleral veins -> ophthalmic veins |

## Nerve Supply of the Eye + Orbit

| Nerve | Type | Supplies |
|---|---|---|
| **CN II (Optic)** | Sensory (special) | Vision — retina to visual cortex |
| **CN III (Oculomotor)** | Motor + Para | SR, IR, MR, IO, LPS + sphincter, ciliary m. |
| **CN IV (Trochlear)** | Motor | Superior oblique ONLY (thinnest CN) |
| **CN VI (Abducens)** | Motor | Lateral rectus ONLY (longest intracranial) |
| **CN V1 (Ophthalmic)** | Sensory | Cornea, conjunctiva, skin of upper lid |
| **Sympathetic** | Autonomic | Dilator pupillae, Muller's muscle, vessels |

★ **CN III Supplies — LR6 SO4 Rest3**
LR = CN VI, SO = CN IV, ALL remaining muscles = CN III (inc. levator palpebrae)

# Orbital Anatomy + Extraocular Muscles

Orbit anatomy + EOM actions = viva favorite! Tables save lives here!

## Walls of the Orbit (Bony)

| Wall | Bones | Clinical Significance |
|---|---|---|
| **Roof** | Frontal + lesser wing sphenoid | Frontal sinus fracture, pulsating proptosis |
| **Floor** | Maxilla + zygomatic + palatine | Blow-out fracture -> infraorbital nerve damage |
| **Medial** | Ethmoid (lamina papyracea) + lacrimal | Thinnest wall, ethmoid sinusitis spread |
| **Lateral** | Zygomatic + greater wing sphenoid | Thickest wall, protects from lateral trauma |

## Important Orbital Openings

| Opening | Contents |
|---|---|
| **Optic canal** | Optic nerve (CN II) + ophthalmic artery |
| **Superior orbital fissure** | CN III, IV, V1, VI + superior ophthalmic vein |
| **Inferior orbital fissure** | Infraorbital nerve + inferior ophthalmic vein |

## Extraocular Muscles (6 muscles)

| Muscle | Primary Action | Nerve | Origin |
|---|---|---|---|
| **Medial Rectus** | Adduction | CN III | Annulus of Zinn |
| **Lateral Rectus** | Abduction | CN VI | Annulus of Zinn |
| **Superior Rectus** | Elevation | CN III | Annulus of Zinn |
| **Inferior Rectus** | Depression | CN III | Annulus of Zinn |
| **Superior Oblique** | Intorsion + depression | CN IV | Body of sphenoid |
| **Inferior Oblique** | Extorsion + elevation | CN III | Maxilla (floor) |

★ **Annulus of Zinn: Common tendinous ring at orbital apex — origin of 4 recti muscles.**
Structures passing THROUGH it: CN III (inf), CN VI, nasociliary nerve, optic nerve.

Anatomy = the base for
EVERY ophthal topic!
You've GOT this Sammy!

## Most Repeated LAQs (10–12 marks)

1. Describe the layers of the eyeball with blood supply and nerve supply
2. Describe anatomy of cornea. Add a note on corneal transparency
3. Write about the arterial supply of the eyeball (diagram essential!)
4. Describe the walls of the orbit with their foramina and contents
5. Describe the anatomy and actions of extraocular muscles

## Most Repeated SAQs (5–6 marks)

1. Layers of cornea with their properties
2. Crystalline lens — anatomy and functions
3. Aqueous humor — formation, circulation, drainage
4. Optic canal and superior orbital fissure contents
5. Annulus of Zinn and structures passing through it
6. Blood supply of retina

## One-Liner Associations (MCQ/Viva Gold)

| Question | Answer |
|---|---|
| Thinnest wall of orbit | Medial (lamina papyracea) |
| Thickest wall of orbit | Lateral wall |
| End artery of eye | Central retinal artery |
| Thinnest part of sclera | Behind recti insertion (0.3mm) |
| Thickest basement membrane | Lens capsule |
| Avascular structures of eye | Cornea, lens, vitreous |
| Refractive power of cornea | +43 D (highest) |
| Refractive power of lens | +15 to +20 D |
| 90% of corneal thickness | Stroma |
| Corneal layer that does NOT regenerate | Bowman's membrane, endothelium |
| Nerve supply of cornea | Long ciliary nerve (CN V1) |
| Muscle NOT from annulus of Zinn | Inferior oblique (from maxilla) |

♥ Anatomy is the foundation, Sammy! Once you know the structure, every disease and surgery will make perfect sense. You're brilliant!